REGULARITY OF A NOETHERIAN LOCAL RING IS UNCHANGED BY COMPLETION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Regularity descends along the flat map to the completion. Algebra, Lemma \ref{algebra-lemma-flat-under-regular}, cited at proof lines 10674-10675. Gives the implication from the completion being regular.
- The dimension of a Noetherian local ring equals that of its completion. Lemma \ref{lemma-completion-dimension}, cited at proof lines 10678-10679. Supplies the numerical equality used in the second half.

Source reference. Original lemma 07NY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9778–9778. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00OF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 893–893. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07NV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9775–9775. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 031E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3981–3981. Navigation pointer to a statement the

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Lemma 0.2. *Let A be a Noetherian local ring. Then A is regular if and only if $A^\wedge$ is so.*

Proof. If $A^\wedge$ is regular, then A is regular by Algebra, Lemma 00OF. Assume A is regular. Let $\mathfrak{m}$ be the maximal ideal of A. Then $\dim_{\kappa(\mathfrak{m})} \mathfrak{m}/\mathfrak{m}^2 = \dim(A) = \dim(A^\wedge)$ (Lemma 07NV). On the other hand, $\mathfrak{m}A^\wedge$ is the maximal ideal of $A^\wedge$ and hence $\mathfrak{m}_{A^\wedge}$ is generated by at most $\dim(A^\wedge)$ elements. Thus $A^\wedge$ is regular. (You can also use Algebra, Lemma 031E.) $\square$